

AWS Agentic AI Architecture Patterns

Building Scalable and Secure AI Agents



AWS AI portfolio

APPLICATIONS AND AGENTS

Kiro

Amazon Quick Suite

AWS Transform

Amazon Connect

AWS Marketplace

FRONTIER AGENTS

Kiro autonomous agent

AWS DevOps Agent

AWS Security Agent

AI & AGENT DEVELOPMENT SOFTWARE & SERVICES

AGENT BUILDERS

Vertically Integrated
Amazon Nova Act

Flexible/OSS
Strands Agents

AMAZON BEDROCK

Models

Amazon Nova

3P Models

Capabilities

Optimization

Customization

Guardrails

Knowledge Bases

AgentCore

Runtime

1P Tools

Gateway

Identity

Memory

Observability

Policy

Evaluations

INFRASTRUCTURE

AMAZON SAGEMAKER

Custom Model Building

Model building

Model training

Deployment

MLOps

Hyperpod

Data Foundation for AI

Data Processing

Storage and Databases

Governance

Vector Search

AI COMPUTE

Trainium

Inferentia

GPUs

INTERFACES & PROTOCOLS
(MCP/A2A)

SECURITY AND POLICIES



Amazon Bedrock: Your Central AI Platform

- **Amazon Bedrock — The Agentic AI Core**
- **Foundation Models** — Access to Amazon Nova and leading third-party models
- **AI Capabilities** — Built-in optimization, guardrails, and customization
- **AgentCore** — Runtime execution, gateway management, memory, identity & observability
- **Knowledge Bases** — Seamlessly ground agents in your enterprise data



Agentic AI Architecture Patterns

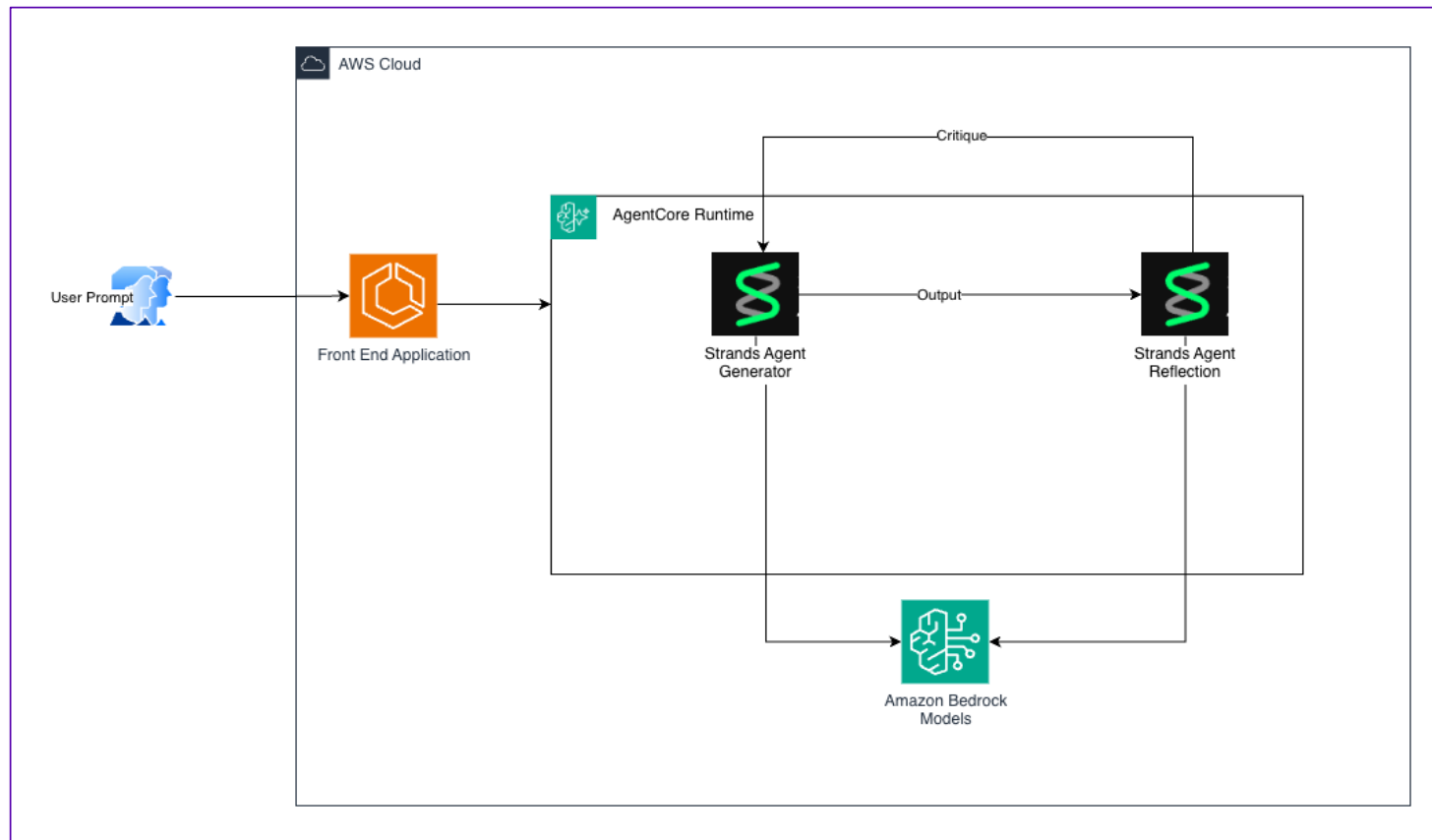


Pattern 1: Reflection Pattern

- Generates initial response
- Evaluates output — reflects and critiques
- Loops back to correct errors and improve quality



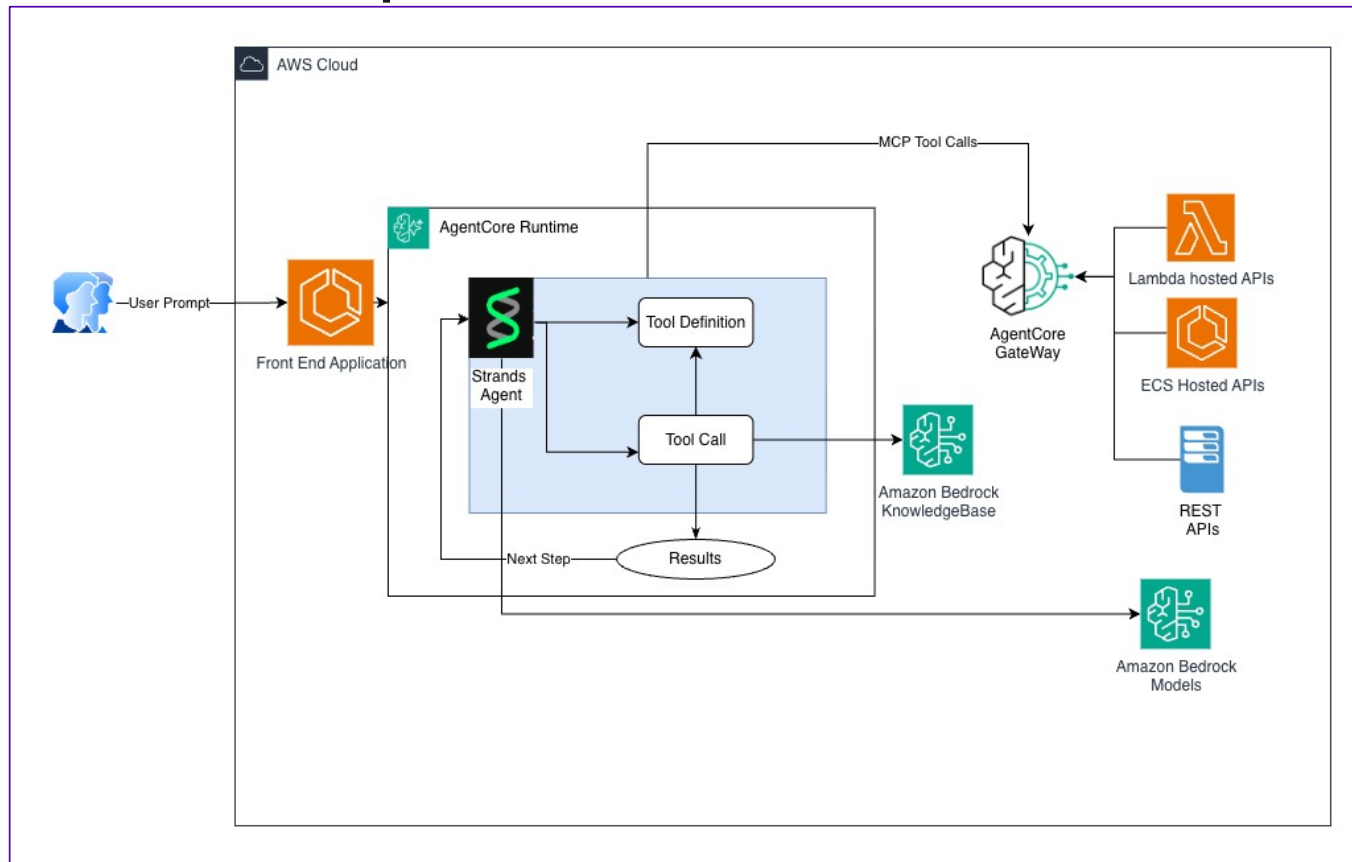
Pattern 1: Reflection pattern



Pattern 2: Tool Use Pattern

- **Tool Use Pattern** *Connect. Retrieve. Act.*
- LLM **dynamically interacts** with the environment at runtime
- Invokes external tools as needed:
 - API calls, RAG / vector store retrieval
 - Browser, ML/LLM models, device control
- **MCP servers** standardize tool interfaces for seamless integration

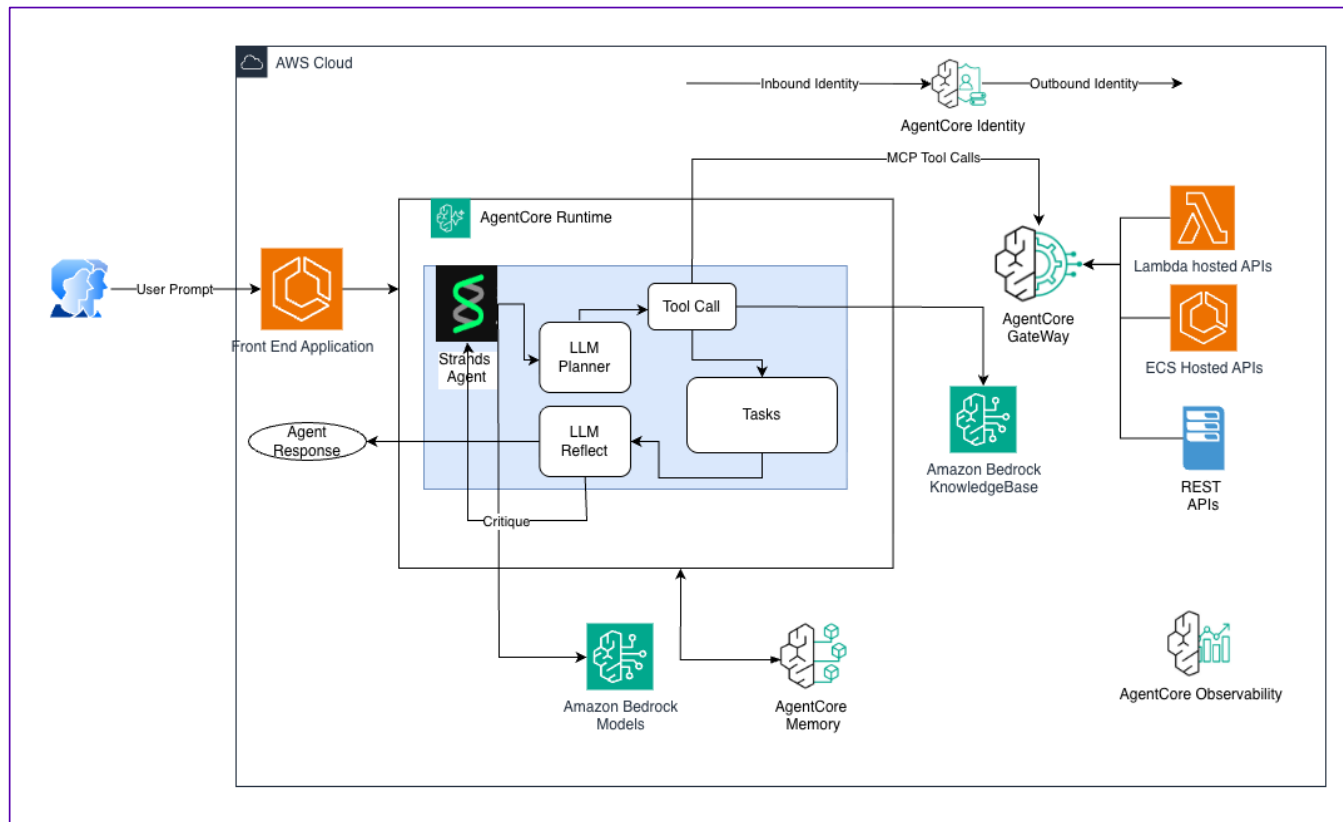
Pattern 2: Tool use pattern



Pattern 3: ReAct - reasoning + tool use pattern

- **ReAct: Reasoning + Action** *Think. Plan. Act.*
- LLM breaks complex tasks into **step-by-step sub-tasks**
- Each step **executes via tools**, feeding results back into reasoning
- Multi-hop reasoning → **dynamic, adaptive responses**

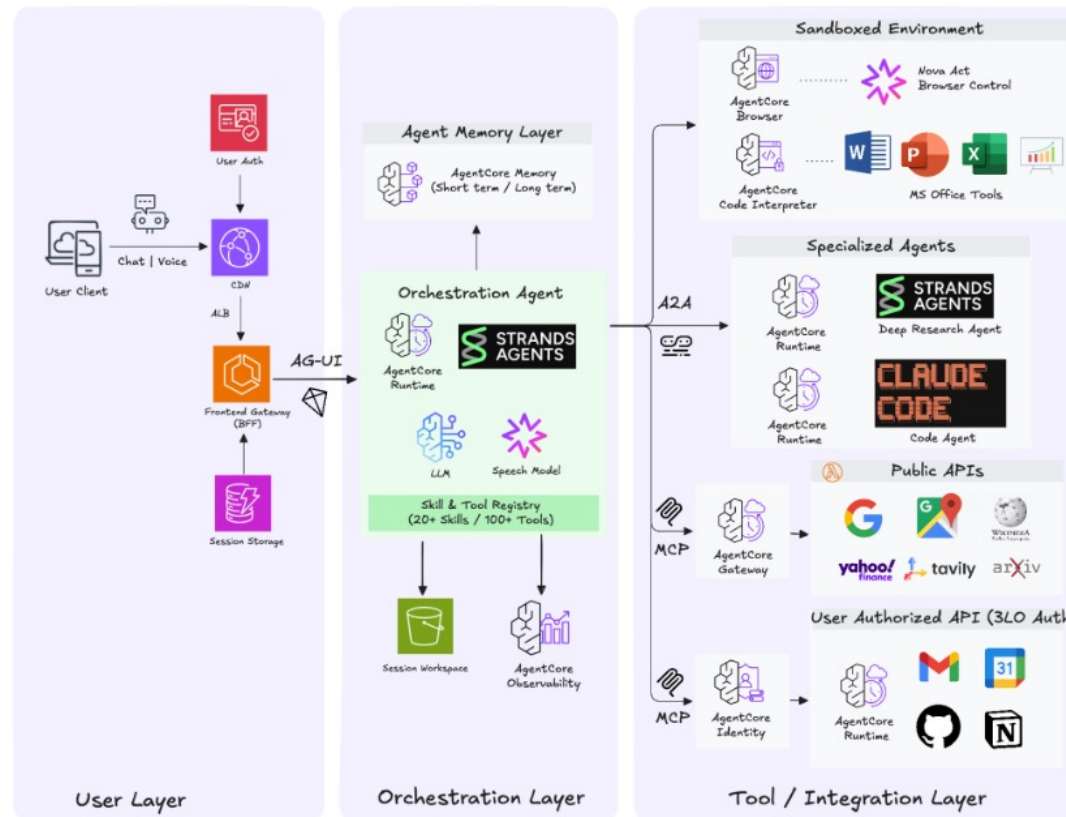
Pattern 3: ReAct - reasoning + tool use pattern



Pattern 4: Planning and orchestration pattern

- **Planning & Orchestration Pattern** *Plan. Delegate. Consolidate.*
- LLM **orchestrator** decomposes the goal into a structured plan of sub-tasks
- Each sub-task is **delegated** to specialized sub-agents or tools for execution
- Orchestrator **collects results, reflects, and re-plans** as needed — then consolidates the final response

Pattern 4: Planning and orchestration pattern

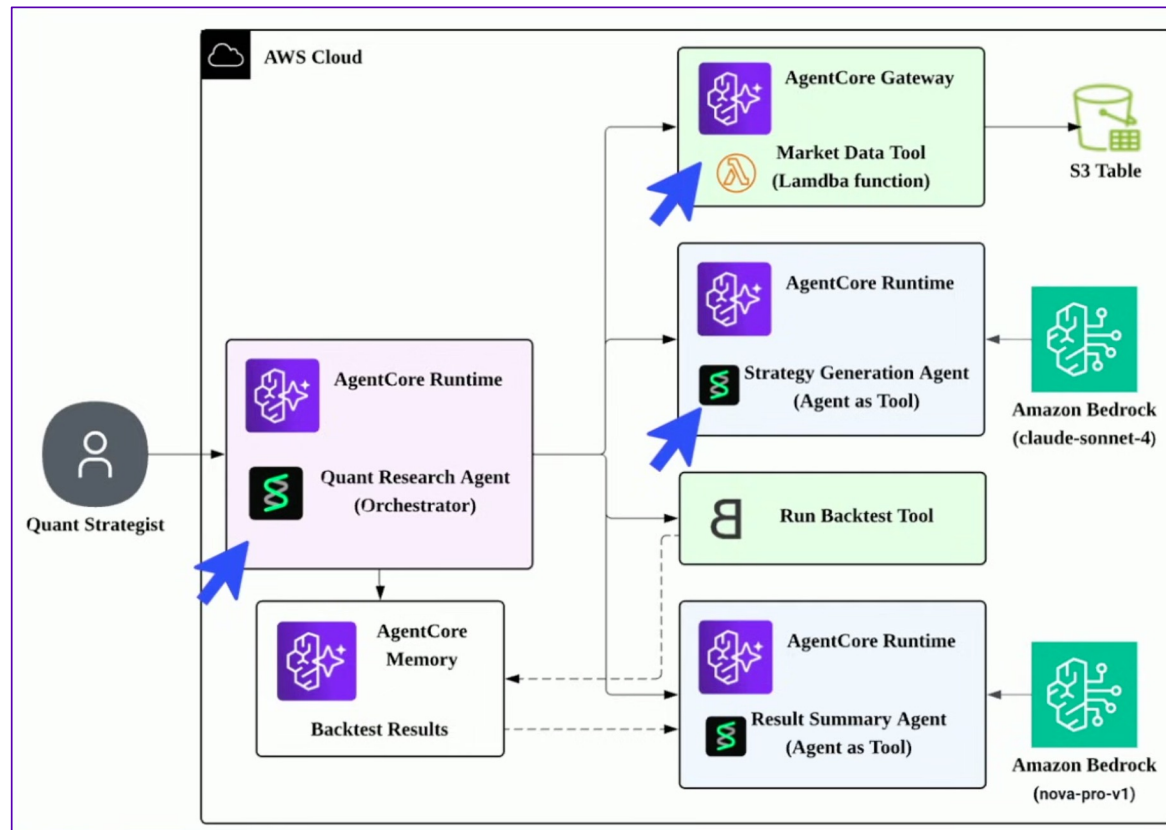


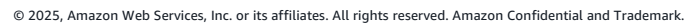
Pattern 5: Multi Agent collaboration Pattern

- Agents as tools
 - specialized AI agents are wrapped as callable functions (tools) that can be used by other agents
- Agent to Agent (A2A) protocol
 - an open standard that defines how AI agents can discover, communicate, and collaborate with each other
- SWARM
 - a collection of autonomous AI agents working together to solve complex problems through collaboration.
- Use case: automated marketing content generation

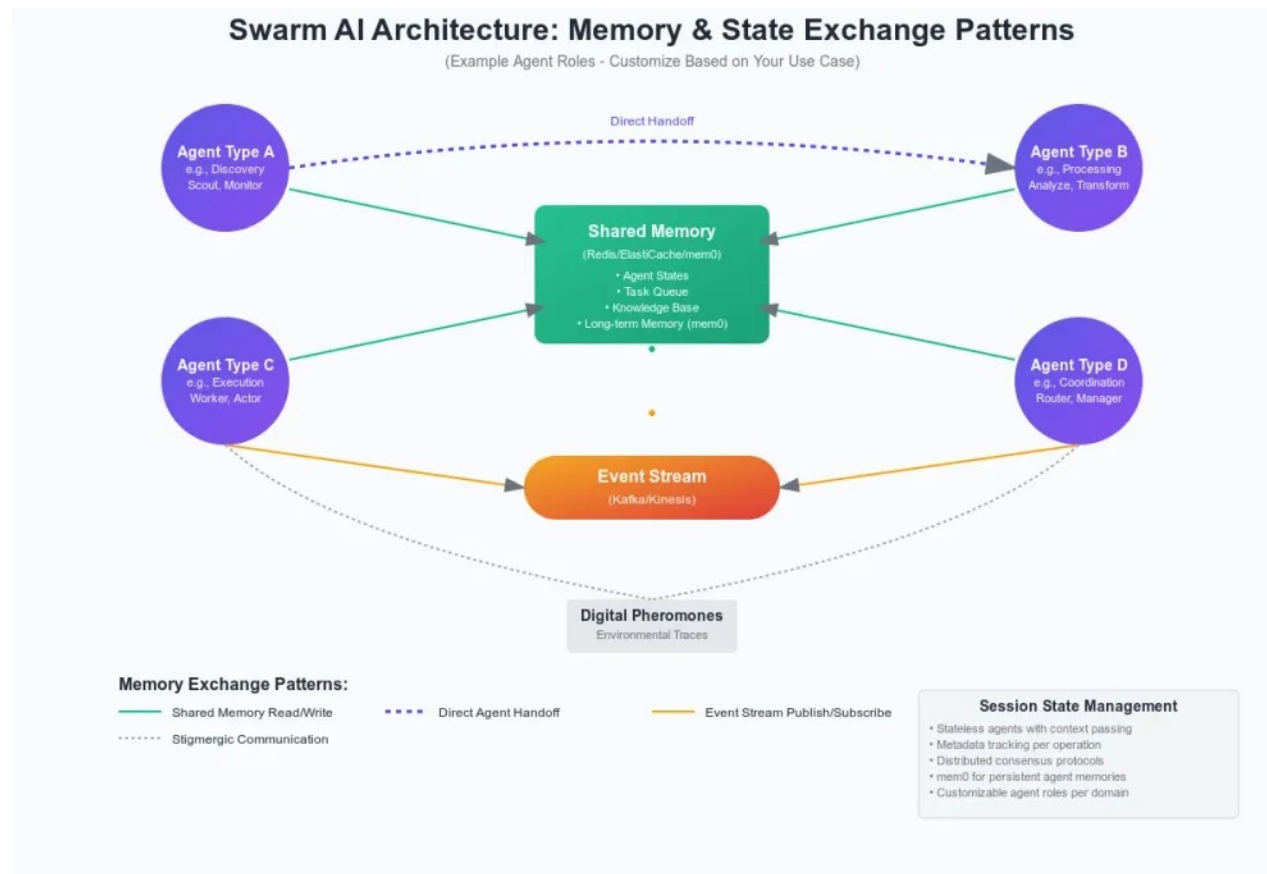


Pattern 5: Multi Agent collaboration Pattern (Agents as Tools)

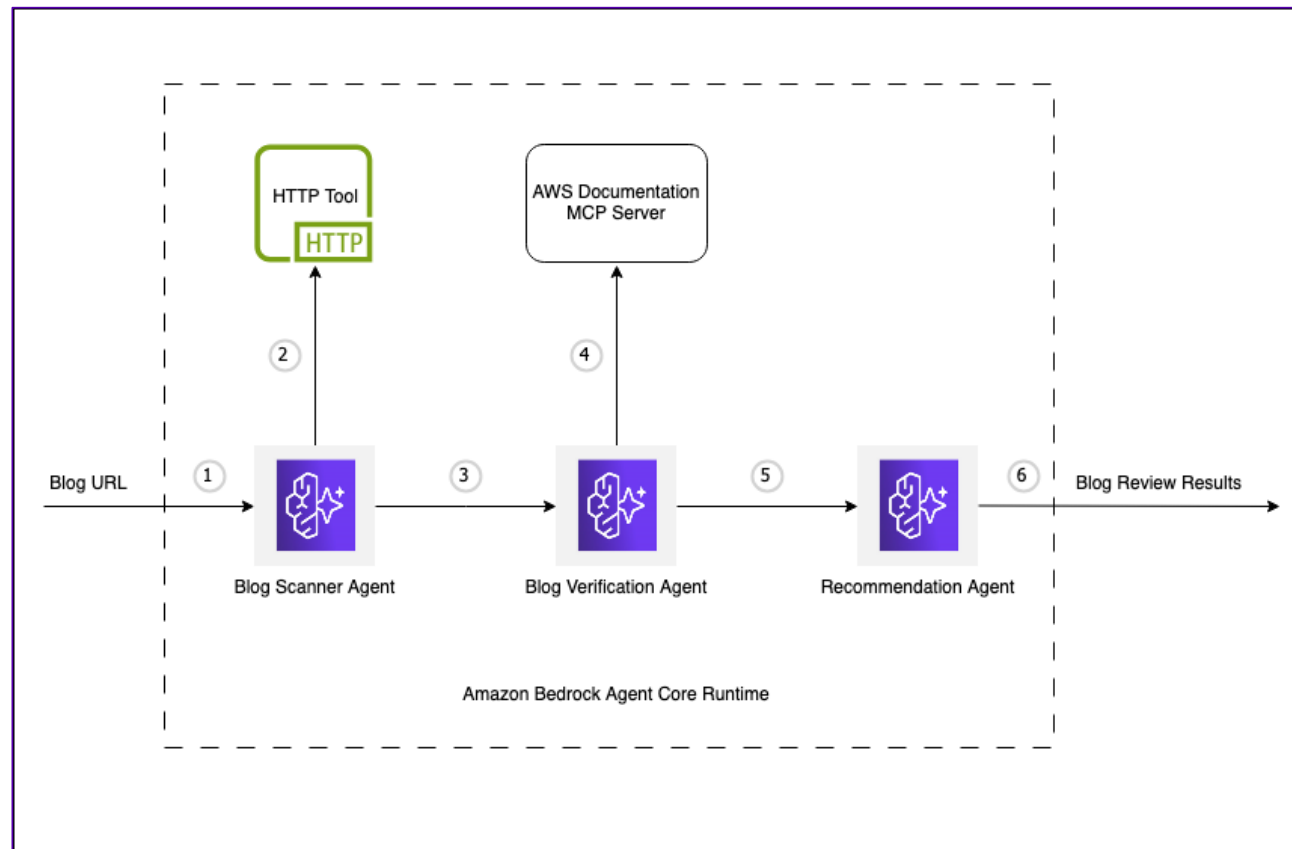




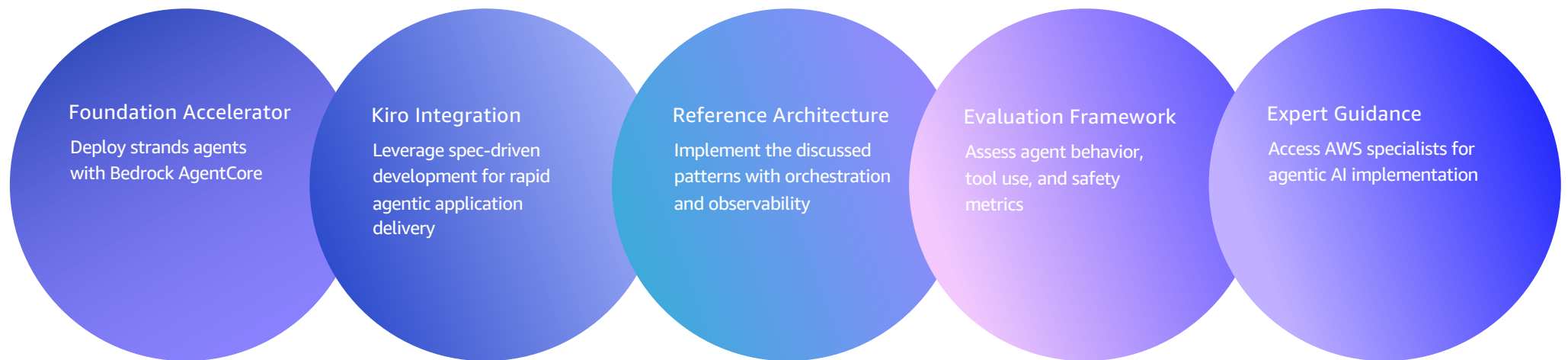
Pattern 5: Multi Agent collaboration Pattern (Swarm)



Pattern 5: Multi Agent collaboration Pattern (Workflow)



Build Production-Ready Agentic AI



Resources

- Reflection pattern - <https://builder.aws.com/content/2zo16pNcEvQHtHpwSaxfFr8nf37/ai-agents-design-patterns-reflection-pattern-using-strands-agents>
- Agents as Tools pattern - <https://aws.amazon.com/blogs/machine-learning/multi-agent-collaboration-patterns-with-strands-agents-and-amazon-nova/>
- Planning and Orchestration Pattern - <https://github.com/aws-samples/sample-strands-agent-with-agentcore/tree/main>
- Graph Pattern - <https://www.youtube.com/watch?v=yxLzAKsAAjo>
- A2A Pattern - <https://aws.amazon.com/blogs/machine-learning/introducing-agent-to-agent-protocol-support-in-amazon-bedrock-agentcore-runtime/>
- Workflow Multi Agent Pattern - <https://aws.amazon.com/blogs/machine-learning/scaling-content-review-operations-with-multi-agent-workflow/>



Thank you!

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